



Guangdong Technion

Israel Institute of Technology

广东以色列理工学院

Calculus 1 - 104003

Midterm
November 29, 2021

Your ID Number:

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First name and Surname:

Guidelines

1. **Duration: 2 hours.** Use of calculators, personal dictionaries, electronic devices, reference materials, personal notes or any other extra material is not allowed.
2. Show all your work. Explain your solutions, quote theorems you are using.
No credit will be given for non-justified answers!.
Write clear and complete answers for each problem in the same page.

1. Consider the function

$$g(x) = \begin{cases} -x & \text{if } x \leq -1 \\ \frac{1-x^2}{1+x} & \text{if } -1 < x \leq 1 \\ \ln x & \text{if } x > 1 \end{cases}$$

- (a) **(7 points)** Find the number or numbers at which g is discontinuous.
- (b) **(7 points)** At which of these numbers g is continuous from the right, from the left or neither?
- (c) **(6 points)** Sketch the graph of g .

Answer

2. **(20 points)** Find an equation for the tangent line to the curve given implicitly by the equation

$$2xy + \pi \sin y = 2\pi$$

at the point $(1, \frac{\pi}{2})$.

Answer

3. Consider the function $f(x) = \frac{x^2-2}{x^2-1}$.
- (a) **(5 points)** Find the largest domain where f is well defined.
 - (b) **(10 points)** Find the limit of f where x tends to infinity/-infinity and find horizontal/vertical/oblique asymptotes (if there are any).
 - (c) **(5 points)** Find the intervals on which f is increasing or decreasing.
 - (d) **(5 points)** Find the local maximum and minimum values of f (if any).
 - (e) **(5 points)** Find the intervals of concavity and the inflection points (if any).
 - (f) **(5 points)** Are there any absolute minimum/maximum values of f ? Explain.
 - (g) **(5 points)** Sketch the graph of f .

Answer

4. Determine whether the following statements are true or false.

In the following questions you need to fill the appropriate empty square which is located to the right of each question.

If you think that statement is False, write the letter **F** inside the square.

If you think that the statement is true, write the letter **T** inside the square.

Recall that you need to prove true statements and you need to provide counter examples for false ones. Show all your work.

- (a) **(5 points)** Let $f(x) = \frac{1-\sqrt{x}}{1-x}$. Then there exists a number $\delta > 0$ such that if $|x - 1| < \delta$ then $|f(x) - \frac{1}{2}| < 0.1$.

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- (b) **(5 points)** The function $f(t) = t^3 - 2t + 2$ has a zero between -2 and 0 .

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- (c) **(5 points)** The function $f(x) = \frac{1}{1+e^{1/x}}$ has a removable (or solvable) discontinuity at $x = 0$.

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- (d) **(5 points)** If f and g are increasing on an interval I , then $f - g$ is increasing on I .

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